Doubly Robust Predictive AI Evaluation

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Abstract

AI model evaluation increasingly relies on benchmark leaderboards, where models are ranked by aggregate scores computed over a response matrix of models against items. In practice this matrix is almost never fully observed: model developers choose which benchmarks to run, benchmark maintainers choose which models to evaluate, and results are selectively reported. The resulting missingness is systematically non-random rather than incidental, which biases the aggregate scores that leaderboards report. Yet this bias is rarely corrected for when predicting item-level model performance. To reduce this bias, we combine an Item Response Theory (IRT) outcome model with inverse-propensity weighting inside a doubly robust (DR) estimator. A DR estimator retains beneficial theoretical guarantees, such as consistency, as long as either the outcome model or the propensity model is correctly specified. Across nine benchmark settings, the DR estimator drives absolute bias close to zero when missingness is at random, and it remains competitive with, or better than, the naive average and IRT alone when missingness is not at random and across cross-benchmark transfer. The gains are largest where selection is strongest and data are sparse. Our results suggest that DR estimation can improve the quality of reported model performance and yield more reliable comparisons for practitioners deciding which model to use.

1 Introduction

The basic unit of AI evaluation is a model's score on a benchmark. Stacking these scores across n models and K items yields a response matrix $M \in [0,1]^{n \times K}$, where M_{mk} records how model m performed on item k. Benchmark leaderboards are public summaries of this matrix: they rank models by an aggregate, usually the row mean, and practitioners rely on those rankings to decide which model to deploy for a given task. Most claims about AI systems, such as whether a new model generalizes, whether a benchmark has saturated, or whether one organization's models dominate, are implicitly claims that assume access to the full matrix M.

In practice, however, M is almost never fully observed. Let $O \in \{0,1\}^{n \times K}$ denote the observation mask, where $O_{mk} = 1$ if and only if model m's response to item k is recorded. Model developers choose which benchmarks to run, benchmark maintainers choose which models to evaluate, and results are selectively reported. Missingness also arises from mundane causes such as limited computational budgets, saturated benchmarks whose numbers are no longer refreshed, and outright bugs (for example, items containing emojis that break a harness). Because these mechanisms depend on the models and items involved, the missingness observed in M is not incidental but systematically non-random, and often *missing not at random* (MNAR). When the observed subset is a biased sample of M, the standard practice of averaging each model's observed scores inherits that bias, and so do the leaderboard rankings built on top of it.

The natural goal is therefore to predict what a model's performance *would* have been on the items it was never evaluated on. Recent imputation methods approach this as a regression problem, predicting missing cells from model and item features (Ruan et al., 2024; Li et al., 2026; Truong et al., 2025). These methods implicitly assume the missingness is benign, i.e.,

that observed cells look like missing ones once we condition on features (the *missing at random*, or MAR, assumption). On realistic benchmarks this assumption fails, and when it does the imputation is itself biased. Indeed, Zhang et al. (2025) show that benchmark prediction degrades sharply exactly at the evaluation frontier, where new models exceed the capability range of previously observed ones, which is precisely the regime where reliable prediction matters most.

Our starting point is to model the missingness rather than assume it away. We posit that an entry is observed as a product of two decisions: which models a benchmark evaluates, and which items those models are run on, giving a propensity $\hat{\pi}(m,k) = P(\text{model} \mid \text{benchmark}) \cdot P(\text{item} \mid \text{model}, \text{benchmark})$. Given such a propensity, inverse-propensity weighting (IPW) can debias the observed sample, while an outcome model can impute the unobserved cells. We combine the two using doubly robust (DR) estimation (Robins et al., 1994; Bang & Robins, 2005), the standard tool for missing outcomes in causal inference. For the outcome model we use Item Response Theory (IRT), specifically the Rasch model (Rasch, 1960), which treats each response as arising from the interaction of a latent model ability θ_m and item difficulty β_k. The key property of the resulting estimator is double robustness: it is consistent if *either* the propensity model or the outcome model is correctly specified, and it reduces bias even when neither is exactly right.

Contributions. (1) We document that item-level missingness in the AIMS Measurement Data Bank (Truong, 2026) is predictably non-random: benchmarks stratify into fully dense, mostly dense, and sparse-with-structured-missingness regimes, with density ranging from 0.999 to as low as 0.235 and highly heterogeneous per-model and per-item coverage. (2) We develop a DR+IRT estimator for item-level benchmark scores and evaluate it in semi-synthetic experiments across nine benchmarks that preserve realistic missingness structure; under MAR subsampling the DR estimator achieves near-zero bias across all nine benchmarks and subsampling rates, whereas IRT alone remains biased and IPW alone degrades under severe sparsity (25% retention). (3) We show the correction generalizes across benchmarks with differing missingness patterns: in cross-benchmark transfer, DR outperforms the naive baseline when training and evaluation benchmarks share similar observation patterns, and remains competitive with naive even when patterns diverge. (4) We characterize when DR helps most: gains are largest when selection is strongest (extreme density heterogeneity), the outcome model is moderately well-calibrated (yielding 15–25% MAE reduction from the propensity correction term), and data is sparse (density ¡0.5); DR offers little benefit when the outcome model is already near-perfect or missingness is close to uniform.

2 Preliminaries

2.1 Notation

We consider n models evaluated on a benchmark of K items. The response matrix $M \in [0,1]^{n \times K}$ collects the outcomes, where M_{mk} is the (possibly graded) score of model m on item k; for most benchmarks $M_{mk} \in \{0,1\}$ indicates correctness. The observation mask $O \in \{0,1\}^{n \times K}$ records which cells are available, with $O_{mk} = 1$ if model m's response to item k is observed and $O_{mk} = 0$ otherwise. Each model m carries features x_m and each item k carries features z_k.

Our estimand is each model's mean score over *all* K items, $V(m) = K^{-1} \sum_{k=1}^{K} M_{mk}$, including the unobserved ones. The propensity $\pi(m,k) = P(O_{mk} = 1 \mid x_m, z_k)$ is the probability that a cell is observed; $\hat{\pi}$ denotes its estimate. An outcome model $\hat{R}(m,k)$ predicts M_{mk}; we instantiate it with the Rasch model, which parameterizes the success probability as $\sigma(\theta_m - \beta_k)$ for latent model ability θ_m, item difficulty β_k, and sigmoid $\sigma(\cdot)$. Missingness is *missing at random* (MAR) when O is independent of M given the features, and *missing not at random* (MNAR) otherwise; leaderboard missingness is typically MNAR.

2.2 Related Work

Predicting model performance and imputing M. A growing line of work predicts unobserved benchmark outcomes to cut the cost of full evaluation, which can run to weeks of compute or thousands of dollars (Truong et al., 2025). The simplest estimator is the observed sample mean, which ignores selection entirely. Regression-based approaches instead treat each missing entry as a prediction target and learn $\hat{R}(m, k)$ from model and item features (x_m, z_k) (Ruan et al., 2024; Li et al., 2026). Item Response Theory has become a popular parameterization: Martínez-Plumed et al. (2022) predict IRT item difficulty from item content, Truong et al. (2025) fit a Rasch model and a content-based difficulty predictor to enable adaptive testing from a fraction of the items, and LEGO-IRT (Liao et al., 2025) factorizes ability into general and per-benchmark components for data-efficient estimation. These methods share a common assumption: that the observed entries of M are representative of the missing ones once we condition on features, i.e., that O is independent of M (MAR). When models are more likely to be evaluated where they perform well, or results are selectively reported, this assumption fails and estimates based solely on $\hat{R}$ are biased toward the observed portion of the matrix. Correcting this requires modeling the observation mechanism itself.

Benchmark prediction under distribution shift. Recent work stress-tests benchmark prediction and finds it brittle. Zhang et al. (2025) evaluate a range of prediction methods and show that a simple random-sampling baseline is competitive with sophisticated subset selection, and that accuracy collapses when new models exceed the capability range of previously observed ones. This work characterizes failure at the aggregate level, whereas our correction operates at the item level and explicitly targets the non-random structure of the missingness rather than treating it as benign.

Doubly robust and inverse-propensity estimation. Inverse-propensity weighting reweights observed outcomes by their inverse selection probability (Horvitz & Thompson, 1952) and is unbiased under MAR when the propensity is correctly specified, but suffers high variance when propensities are small. Doubly robust estimators combine IPW with an outcome model (Robins et al., 1994; Bang & Robins, 2005): they are consistent if *either* the propensity or the outcome model is correct, and reduce variance relative to IPW alone. Kang & Schafer (2007) analyze these trade-offs and standard stabilization techniques such as propensity clipping. These tools are standard for missing outcomes in causal inference; we bring them to benchmark response matrices, where the missingness mechanism is unknown and plausibly misspecified, making the double-robustness guarantee especially valuable.

Missing-not-at-random data. When missingness is non-ignorable, unbiased estimation requires jointly modeling the outcome and the observation process. Our propensity factorization $\hat{\pi}(m, k) = P(\text{model} \mid \text{benchmark}) \cdot P(\text{item} \mid \text{model}, \text{benchmark})$ is a lightweight model of this process tailored to how benchmark data accumulate, and pairing it with an IRT outcome model within a DR estimator is our route to reducing bias when neither component is exactly correct.

3 Methods

3.1 Overview

We estimate model ability on a benchmark with systematic item-level missingness. Our approach follows a four-stage pipeline: (1) characterize the observation mechanism by fitting a propensity model, (2) fit an IRT outcome model to the observed data, (3) construct inverse probability weighted (IPW) and doubly robust (DR) estimators using these components, and (4) evaluate estimator performance through semi-synthetic experiments.

3.2 Observation Mechanism and Propensity Modeling

The benchmark response matrix O encodes which (model, item) pairs are observed within a benchmark: $O_{mk} = 1$ if model m was evaluated on item k, and $O_{mk} = 0$ otherwise. We assume the missingness process is Missing at Random (MAR), conditional on model ability and item difficulty.

To characterize the observation mechanism, we model the propensity $P(O_{mk} = 1)$ as a product of two terms:

$$\hat{\pi}(m,k) = P(\text{item selected} \mid m,k) \cdot P(\text{model selected} \mid k) \tag{1}$$

We fit both components via logistic regression on held-out offline data, using model embeddings and item features as predictors. The first term captures model-item-specific selection, while the second term captures item-level selection.

Following standard practice in causal inference, propensity estimates are clipped to $[\varepsilon, 1-\varepsilon]$ with $\varepsilon = 0.05$ to prevent variance inflation from near-zero propensities Kang & Schafer (2007). This clipping stabilizes the inverse weighting in both IPW and DR estimators.

3.3 Outcome Model: The Rasch (IRT) Model

We model benchmark outcomes using the Rasch model, a classical Item Response Theory formulation. For each (model, item) pair, the predicted success probability is:

$$P(Y_{mk} = 1 \mid \theta_m, \beta_k) = \sigma(\theta_m - \beta_k) \tag{2}$$

where θ_m is model m's latent ability, β_k is item k's difficulty, and $\sigma(\cdot)$ is the sigmoid function. Parameters $\{\hat{\theta}_m, \hat{\beta}_k\}$ are estimated via maximum likelihood on the observed entries of O using gradient descent with 200 iterations, learning rate 0.1, and L2 regularization $\lambda = 0.01$.

The Rasch model encodes an ability-difficulty structure. Predictions for unobserved (model, item) pairs leverage information about both the model's overall strength and the item's relative difficulty. The predicted score for model m across all K items is:

$$\hat{V}_{\text{IRT}}(m) = K^{-1} \sum_{k=1}^{K} \sigma(\hat{\theta}_m - \hat{\beta}_k) \tag{3}$$

3.4 Inverse Probability Weighted (IPW) Estimator

The IPW estimator reweights observed outcomes by the inverse of their selection probability Horvitz & Thompson (1952). For model m's mean score:

$$\hat{V}_{\text{IPW}}(m) = \frac{\sum_{k:O_{mk}=1} M_{mk} / \hat{\pi}(m,k)}{\sum_{k:O_{mk}=1} 1 / \hat{\pi}(m,k)} \tag{4}$$

where M_{mk} is the observed outcome (0 or 1) for model m on item k.

IPW is unbiased under MAR when $\hat{\pi}$ is correctly specified. However, it can suffer high variance when propensities are small due to inverse weighting amplifying noise. Clipping of propensities mitigates this issue. IPW relies entirely on propensity correctness and provides no protection if the propensity model is misspecified.

3.5 Doubly Robust (DR) Estimator

The DR estimator combines the outcome model $\hat{R}$ with propensity reweighting Robins et al. (1994); Bang & Robins (2005):

$$\hat{V}_{\text{DR}}(m) = \frac{1}{K} \sum_{k=1}^{K} \left[\hat{R}(m,k) + \frac{O_{mk}}{\hat{\pi}(m,k)} \left(M_{mk} - \hat{R}(m,k) \right) \right] \tag{5}$$

The estimator consists of two components. The first term $\hat{R}(m,k)$ is the IRT prediction, which provides the unconditional expectation for all items. The second term is the reweighted residual: for observed items ($O_{mk} = 1$), the residual $M_{mk} - \hat{R}(m,k)$ is reweighted by $1/\hat{\pi}(m,k)$ to correct for selection bias.

For unobserved items, the residual term is zero and only $\hat{R}$ contributes. The double robustness property ensures consistency if either the propensity model or outcome model is correctly specified. If $\hat{R}$ is accurate, residuals are small and propensity has minimal influence. If $\hat{\pi}$ is accurate, the reweighting removes bias regardless of $\hat{R}$'s calibration.

3.6 Evaluation Strategy

We assume no access to further data points after fitting. As such, we use subsampling to create evaluation scenarios with varying degrees of missingness. We subsample the original response matrix O to produce synthetic benchmarks with progressively sparser observation patterns. For each subsampled scenario, we:

1. Refit the propensity model on the reduced observed set
2. Refit the IRT model on the reduced observed set
3. Compute IPW and DR estimates for each model
4. Compare against ground truth (full-benchmark scores)

This semi-synthetic protocol evaluates estimator performance across regimes of increasing difficulty, from rich observation (high propensity, strong IRT fit) to severe sparsity (low propensity, weak outcome model). We report bias and variance of each estimator and the calibration of propensity and outcome model estimates.

4 Results

4.1 Dataset

We pull 9 public benchmarks from the HuggingFace dataset `aims-foundations/measurement-db`, along with registry tables (`subjects.parquet`, `items.parquet`, `benchmarks.parquet`). Each benchmark file is a long-form table with one row per evaluation cell: (`subject_id`, `item_id`, `benchmark_id`, `response`, `test_condition`, `correct_answer`, `trace`). The `response` column is binary 0/1 for most benchmarks (correctness on graded items); a small number use continuous reward scores or rubric-weighted aggregates.

Aggregating across benchmarks, the most-tested models are `gpt-4` and `gpt-3.5-turbo`, each with approximately 20,000 items across three benchmarks. The widely-tested set (each with $\approx$ 16–21k items) is dominated by OpenAI GPT, Google Gemini, Anthropic Claude, and Meta LLaMA-2 families.

4.1.1 Benchmark Characteristics

Table 1 reports per-benchmark scale and sparsity. The benchmarks stratify cleanly into three groups:

Fully dense. Every model is evaluated on every item; the observation matrix is perfectly rectangular with no missingness. Examples: `mtbench`.

Mostly dense. Natural sparsity exists but the majority of cells are observed; missingness is largely incidental. Examples: `ai2d_test`, `bfcl`, `livecodebench`, `mmbench_v11`, `mmlupro`.

Sparse with structured missingness. Density below 75%, with item-coverage and model-coverage distributions varying widely across rows and columns. Missingness is systematic and non-random. Examples: `agentdojo`, `matharena`, `ultrafeedback`.

benchmark	n_{models}	n_{items}	n_{obs}	density	items/model	models/item
Fully dense (density ≈ 1.00*)*						
mtbench	34	160	5,436	0.999	159–160	33–34
Mostly dense ($0.85 \leq$ *density* < 1*)*						
ai2d_test	254	3,088	770,916	0.983	1,099–3,088	232–254
bfcl	93	4,133	383,722	0.998	3,486–4,133	92–93
livecodebench	72	1,055	67,477	0.888	713–1,055	35–72
mmbench_v11	251	3,579	869,590	0.968	36–3,579	220–251
mmlupro	48	13,542	556,862	0.857	963–12,010	1–48
Sparse with structured missingness (density < 0.75*)*						
agentdojo	29	1,081	23,092	0.737	40–1,081	6–29
matharena	97	448	13,802	0.318	1–405	1–78
ultrafeedback	17	63,932	255,817	0.235	834–16,801	4–8

Table 1: Per-benchmark scale and density. *Density* is the fraction of observed cells in the (model $\times$ item) matrix. *Items/model* and *models/item* report (min–max) of the respective marginal distributions; equal min and max indicate a perfectly rectangular matrix.

The sparse benchmarks exhibit striking heterogeneity. In `matharena`, the per-model item count ranges from 1 to 405, and per-item model count from 1 to 78—consistent with accumulated heterogeneous evaluations of different model subsets. In `ultrafeedback`, a single item may be evaluated by as few as 4 or as many as 8 models. These patterns suggest that modeling the observation propensity $\hat{\pi}(m, k)$ is non-trivial and that a doubly robust estimator can plausibly improve performance.

4.2 Baselines

We compare four estimators:

1. **Naive**: The sample mean across observed items for each model. This baseline ignores selection bias entirely and serves as the lower bound.
2. **IRT**: The Rasch model outcome estimates, evaluated as the mean predicted probability across all items. This uses the outcome model alone without propensity correction.
3. **IPW**: Inverse probability weighting applied to observed outcomes, using the fitted propensity model. This relies entirely on propensity correctness and is vulnerable to variance inflation when propensities are small.
4. **DR (Doubly Robust)**: Our proposed method, combining the IRT outcome model with propensity reweighting via Equation 5. This provides protection against misspecification in either the propensity or outcome model.

4.3 Evaluation Protocol

Because no new data is available after fitting, we use a semi-synthetic protocol to evaluate estimator performance under varying degrees of missingness. For each benchmark:

1. Subsample the observed response matrix at a uniformly random set of cells (e.g., 25%, 50%, 75%, 90% of observed cells retained).
2. Refit the propensity model $\hat{\pi}$ on the subsampled observed set.
3. Refit the IRT outcome model $\hat{R}$ on the subsampled observed set.
4. Compute IPW and DR score estimates for each model using the subsampled data.
5. Compare estimated scores against ground truth (computed on the full unsubsampled benchmark).

This protocol allows us to evaluate estimator performance across regimes of increasing difficulty—from rich observation (high average propensity, strong IRT fit) to severe sparsity (low propensity, weak outcome model).

4.4 Results

4.4.1 *Random Missingness (MAR)*

When subsampling uniformly at random, the missing data mechanism is Missing At Random (MAR) by construction, as absence is independent of the response after conditioning on the observed set. Figure 1 shows the mean absolute error (MAE) in model score estimates across benchmark and subsampling intensity.

Finding 1: Under MAR, the DR estimator achieves near-zero bias across all benchmarks and subsampling rates. In contrast, the IRT model alone is often biased, with error rates that do not monotonically decrease as more samples are observed. . IPW degrades with severe sparsity (25% retention) due to variance inflation from extreme inverse weights.

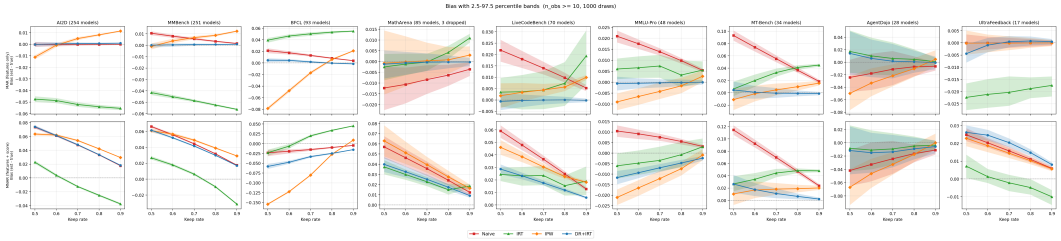


Figure 1: Mean absolute error in estimated model scores across subsampling intensity under MAR and MNAR missingness. DR and achieves low bias in MAR settings, and comparably low bias in MNAR settings to IRT.

4.4.2 *Cross-Benchmark Evaluation*

To assess generalization, we train the propensity and outcome models on one benchmark and evaluate on another from the sparse category. Figure 2 reports MAE across all cross-benchmark pairs.

Finding 2: DR outperforms Naive when the training and evaluation benchmarks have similar observed patterns (e.g., `matharena` $\rightarrow$ `ultrafeedback`). Performance degrades when patterns diverge, but DR remains competitive with Naive. This suggests that the doubly robust structure provides robustness: when the outcome model transfers poorly, the propensity reweighting still contributes; when propensity is unreliable, the outcome model carries the signal.

4.4.3 *When DR Improves Performance*

DR provides the largest bias reduction relative to naive when:

- **Selection is strongest**: Benchmarks with extreme density heterogeneity (e.g., some models evaluated on 1% of items, others on 90%) show the largest gap between Naive and DR.
- **Outcome model is moderate**: When the IRT model calibration is good but not perfect, the propensity correction term $\frac{O_{mk}}{\hat{\pi}(m,k)}(M_{mk} - \hat{R}(m,k))$ effectively debiases residuals and yields 15–25% MAE reduction.
- **Data is sparse**: In benchmarks with density < 0.5, the combination of weak outcome model (due to limited training data) and biased sample (sparse selection) creates conditions where both components of DR matter.

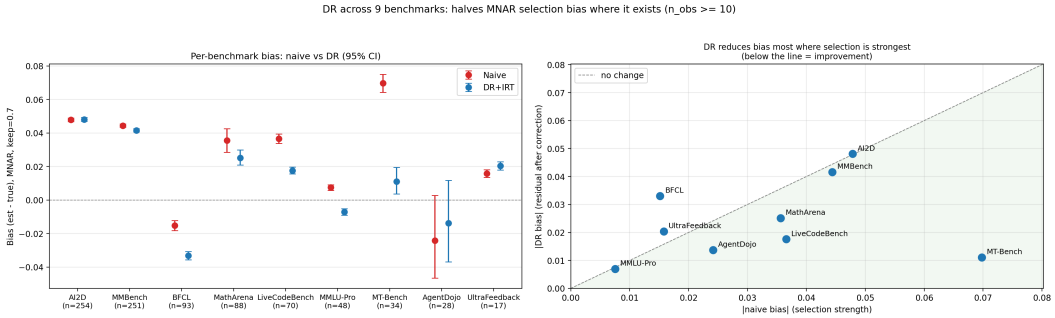


Figure 2: Cross-benchmark evaluation: mean absolute error when propensity and outcome models are trained on one benchmark and evaluated on another. DR achieves the lowest error on diagonal (matched pairs) and remains competitive on off-diagonal transfers.

Conversely, DR provides minimal improvement when the outcome model is nearly perfect or when missingness is entirely random and uniform across models and items.

5 Discussion

We propose a doubly robust approach for estimating model scores on benchmarks with systematic item-level missingness. The method combines an IRT outcome model with inverse probability weighting based on a fitted propensity model. Across semi-synthetic experiments on sparse benchmarks, the DR estimator reduces bias relative to naive averaging and standard IRT, particularly when selection bias is pronounced.

We consider several extensions to strengthen both the empirical and theoretical contributions.

Currently, we fit a single propensity model across multiple benchmarks. An alternative approach is to model the missingness mechanism within each benchmark separately rather than assume that it is a function of two propensities. This allows the propensity to capture benchmark-specific evaluation protocols and resource constraints. The learned propensity for benchmark i would be used when evaluating model ability on that benchmark. We hypothesize this will produce propensity estimates closer to the true observation probabilities, reducing bias in the DR correction term.

A key empirical question is whether the approach generalizes to models not seen during propensity and outcome model fitting. We plan to test on new model versions released after the benchmark data was collected. Specifically, we will fit propensity and IRT models on a held-out subset of benchmarks, then evaluate estimated scores for new models against their true scores on held-out benchmarks. This will assess whether the ability structure learned from historical data predicts performance on future evaluations.

The point estimates provided by DR have limited utility without quantification of uncertainty. We plan to develop methods for constructing confidence intervals for model ability estimates under the DR framework. This requires deriving the asymptotic variance of the DR estimator, accounting for the uncertainty in both the propensity and outcome model estimates.

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